

Project: BladeCore

M54C Evaluation Board

Rev 1.0.0

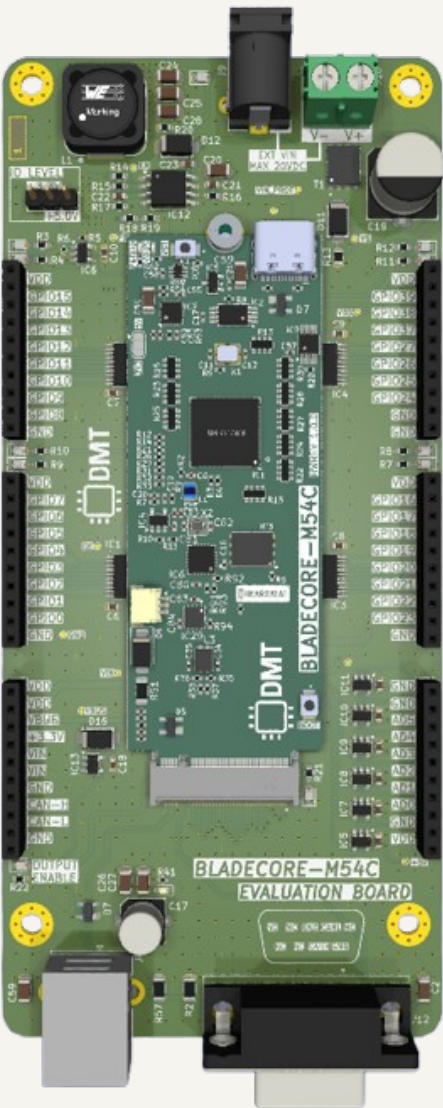
21-03-2026

CONTENT

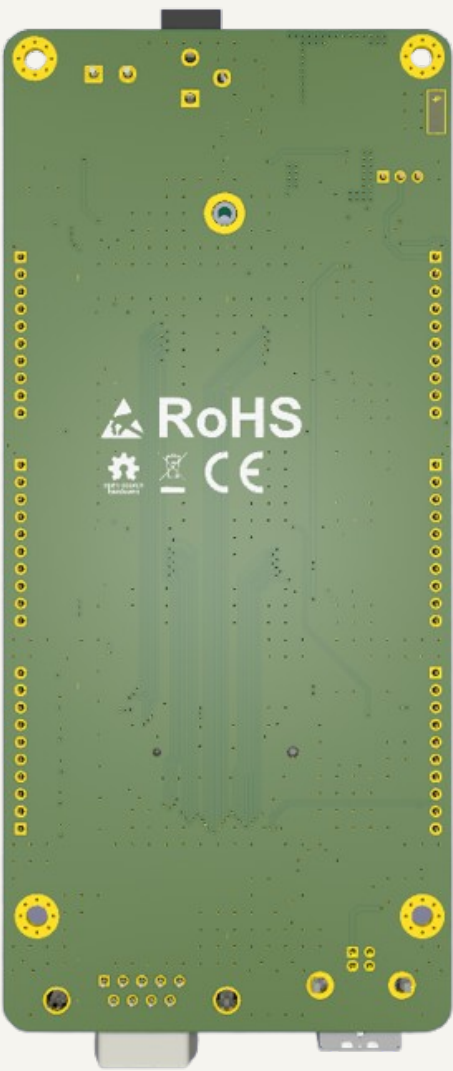
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PCB PREVIEW

TOP VIEW



BOTTOM VIEW



COMMENT GUIDELINES

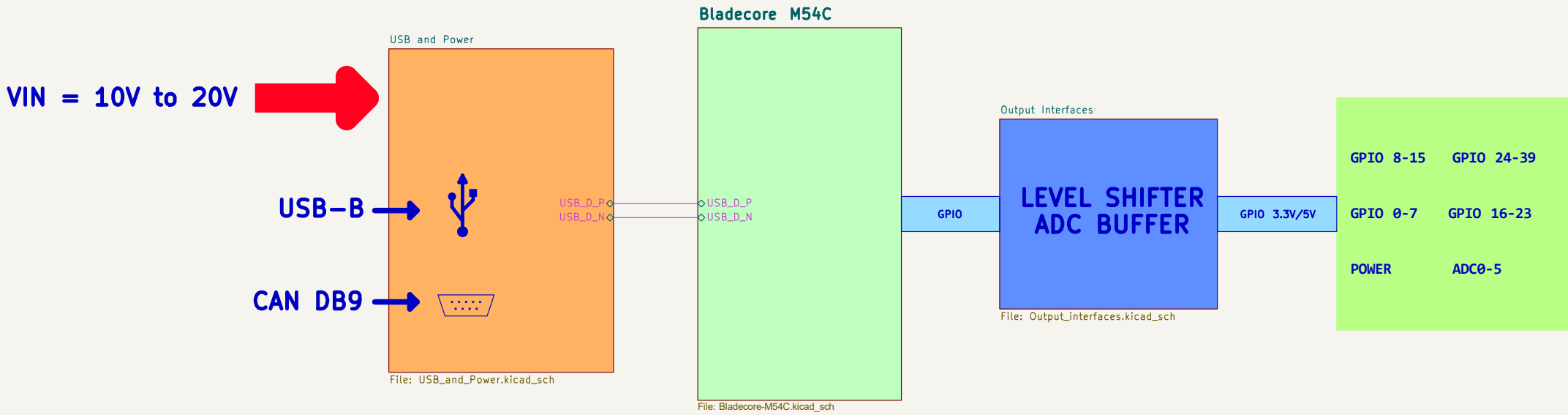
General comments are black, 50 mil size
Design notes and guidelines are blue, 50 mil size
Layout instructions are red, 50 mil size

NOTES

Not fitted components are marked as **X**

 Date: 2025-01-12 Sheet Title: M54-DevBoard_1.0.0	Board Name: BladeCore		Project Name: M54C Evaluation Board	
	File Name: M54-DevBoard_1.0.0.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 1 of 6

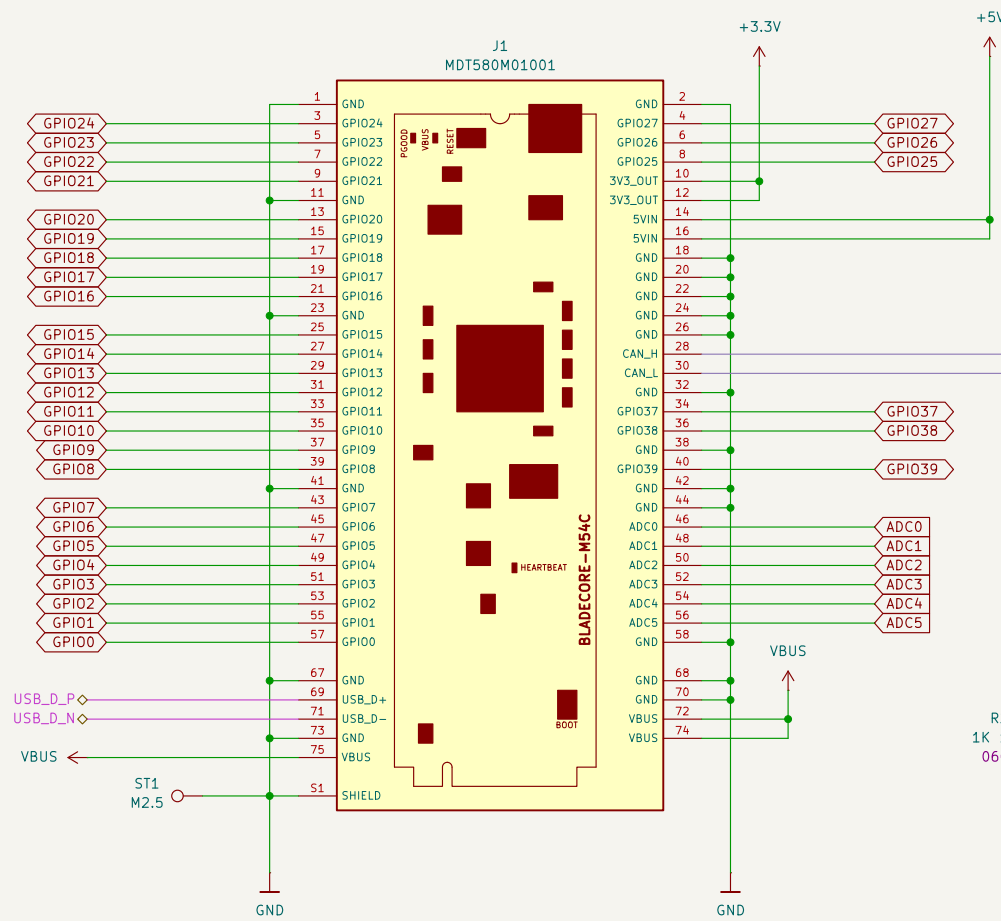
[2] Project Architecture



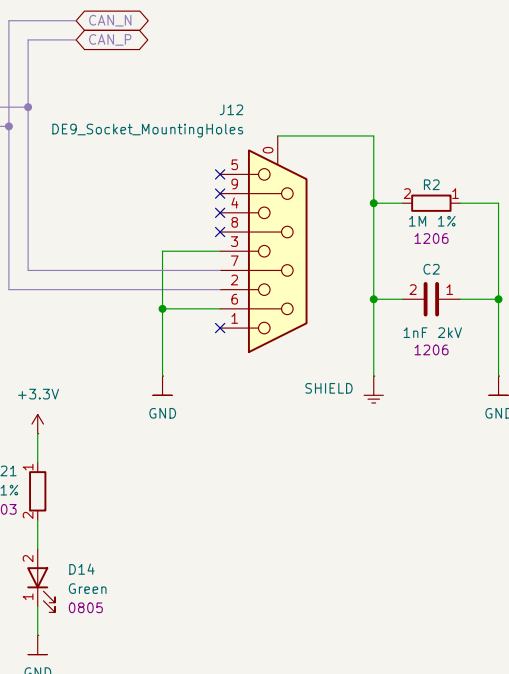
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	File Name: Project Architecture.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 2 of 6

[3] Sheet Title A

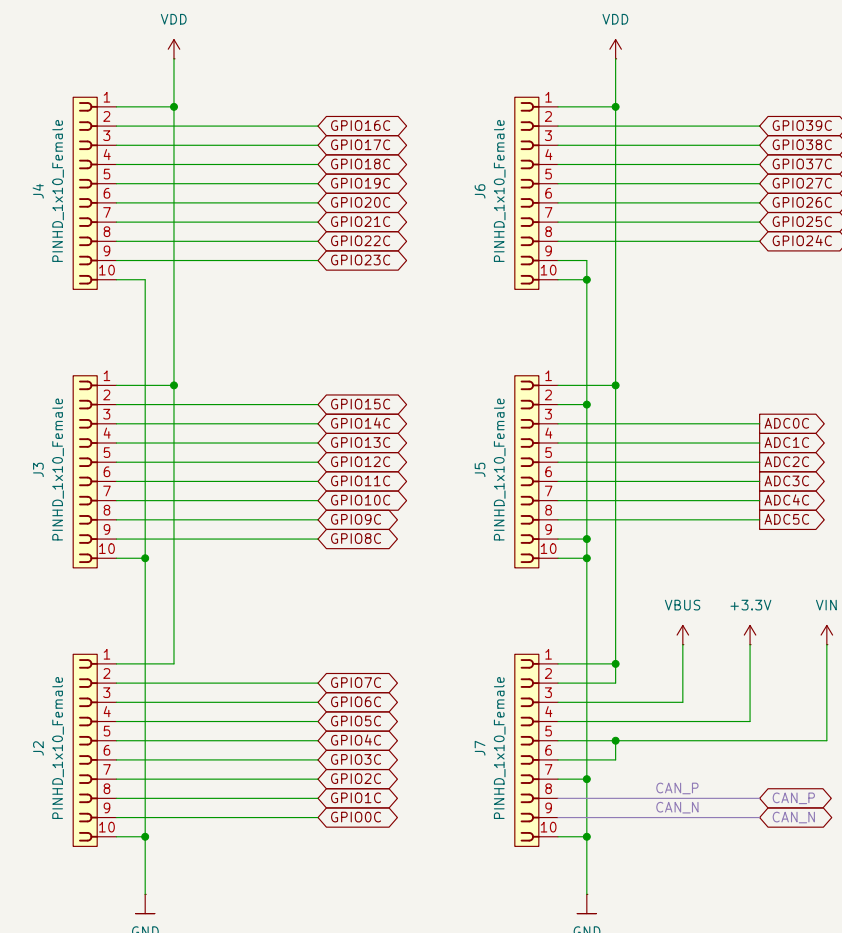
BLADECORE-M54C SOCKET



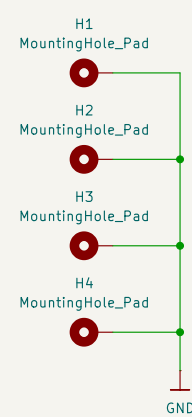
CAN DB9 CONNECTOR



GPIO HEADERS

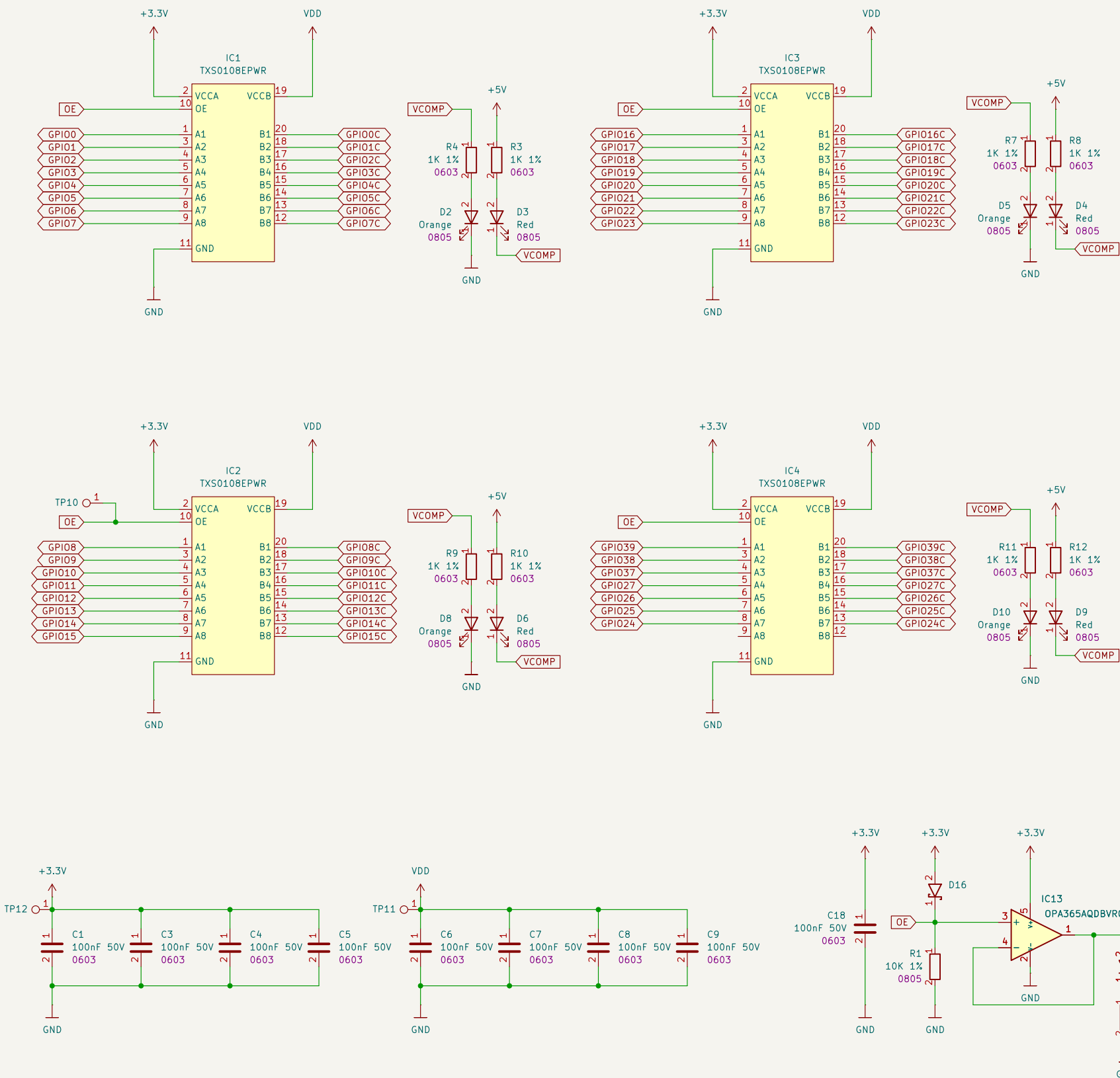


MOUNTING HOLES

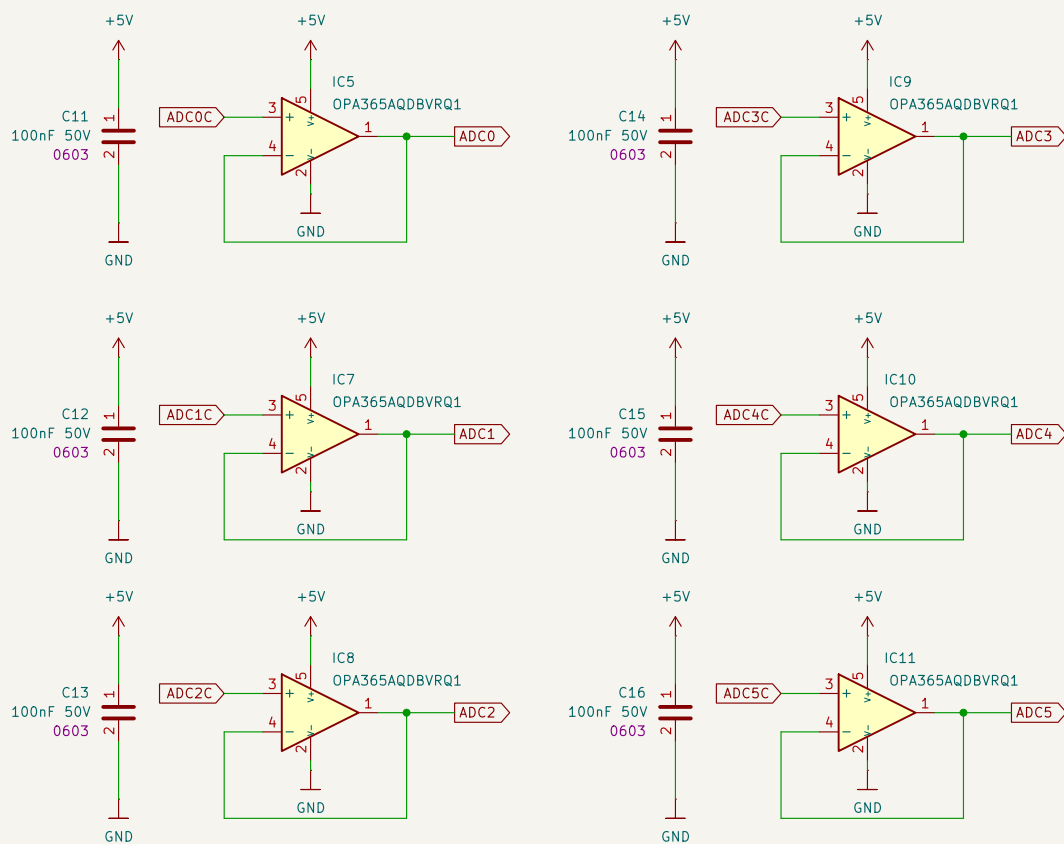


 Date: 2025-01-12 Sheet Title: Bladecore M54C	Board Name: BladeCore		Project Name: M54C Evaluation Board	
	File Name: Bladecore-M54C.kicad_sch		Revision: 1.0.0	Variant:
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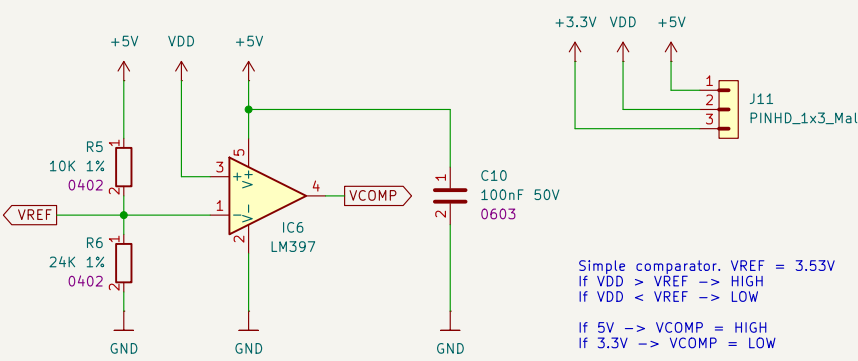
GPIO LEVEL SHIFTERS



ADC INPUT BUFFERS



VDD LEVEL SELECTOR AND COMPARATOR



 Date: 2025-01-12 Sheet Title: Output Interfaces	Board Name: BladeCore File Name: Output_interfaces.kicad_sch		Project Name: M54C Evaluation Board Revision: 1.0.0	
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 4 of 6

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INPUT SUPPLY WITH REVERSE POLARITY PROTECTION

10V-36V to 5V 3A BUCK

This schematic diagram illustrates the input supply and buck converter circuit. The input is connected through a connector J9 (PJ-102AH) to a network of components including a diode D11 (SMAZ15-13-F) for reverse polarity protection, a large electrolytic capacitor C19 (330uF 50V), and several smaller capacitors (C20, C21, C22, C23) and resistors (R13, R14, R15, R16, R17, R18, R19, R20). The circuit is powered by a buck converter IC12 (MP4462DN-LF-Z). The output of the buck converter is a +5V supply, which is filtered by a network of capacitors (C24, C25, C28) and a diode D13 (CM5H3-40MA or B340A-13-F). The output is connected to a terminal TP4.

USB CONNECTOR

This schematic diagram shows the USB connector circuit. It features a USB connector J8 (USB-B-S-F-B-TH) connected to a network of components including a diode D1 (Blue 0805) for reverse polarity protection, a large electrolytic capacitor C17 (100uF 50V), and several smaller capacitors (C26, C27, C29) and resistors (R41, R57). The circuit is powered by a USB connector IC12 (SP0503BAHTG). The output of the USB connector is a +5V supply, which is filtered by a network of capacitors (C24, C25, C28) and a diode D13 (CM5H3-40MA or B340A-13-F). The output is connected to a terminal TP4.

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INPUT SUPPLY WITH REVERSE POLARITY PROTECTION

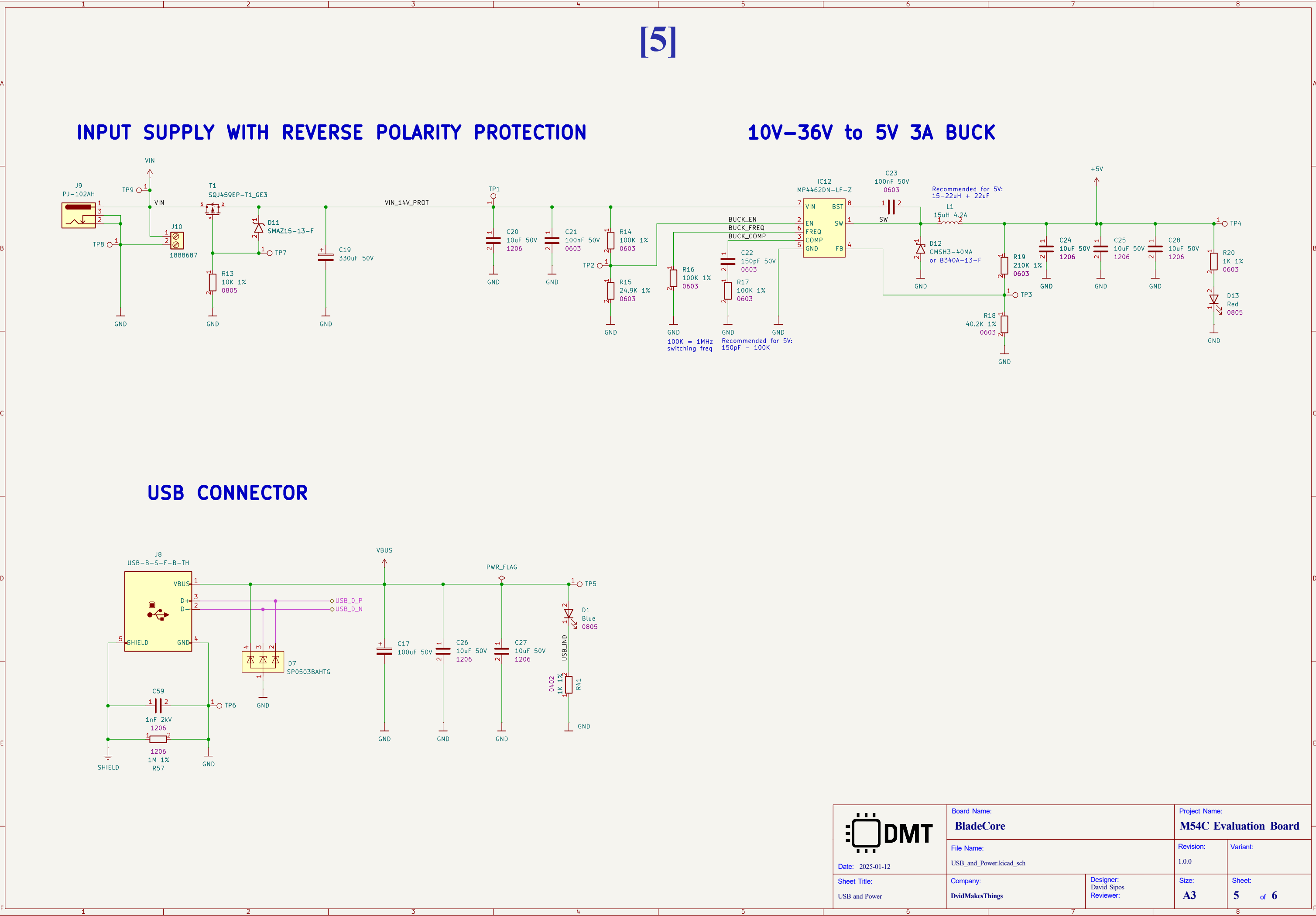
10V-36V to 5V 3A BUCK

This schematic diagram illustrates the input supply and buck converter circuit. The input is connected through a connector J9 (PJ-102AH) to a reverse polarity protection circuit consisting of a diode D11 (SMAZ15-13-F) and a MOSFET T1 (SQJ459EP-T1_GE3). The input voltage is labeled VIN, and the output of the protection circuit is labeled VIN_14V_PROT. The circuit includes several test points (TP8, TP9, TP7, TP1, TP2, TP3, TP4) and various passive components: resistors R13 (10K 1% 0805), R14 (100K 1% 0603), R15 (24.9K 1% 0603), R16 (100K 1% 0603), R17 (100K 1% 0603), R18 (40.2K 1% 0603), R19 (210K 1% 0603), R20 (1K 1% 0603), and R57 (1M 1% 1206). Capacitors include C19 (330uF 50V), C20 (10uF 50V 1206), C21 (100nF 50V 0603), C22 (150pF 50V 0603), C23 (100nF 50V 0603), C24 (10uF 50V 1206), C25 (10uF 50V 1206), C26 (10uF 50V 1206), C27 (10uF 50V 1206), C28 (10uF 50V 1206), and C59 (1nF 2kV 1206). The buck converter is implemented using the IC12 (MP4462DN-LF-Z) with pins VIN, EN, FREQ, COMP, GND, SW, and FB. The output of the buck converter is +5V, which is filtered by a capacitor C28 and a diode D12 (CMSH3-40MA or B340A-13-F). The output voltage is labeled +5V, and the output current is 3A. The circuit also includes a diode D13 (Red 0805) and a resistor R20 (1K 1% 0603) for output regulation.

USB CONNECTOR

This schematic diagram shows the USB connector circuit. The USB connector is J8 (USB-B-S-F-B-TH) with pins VBUS, D+, D-, SHIELD, and GND. The VBUS line is connected to a diode D1 (Blue 0805) and a resistor R41 (1K 1% 0402). The D+ and D- lines are connected to a diode D7 (SP0503BAHTG) and a resistor R57 (1M 1% 1206). The SHIELD line is connected to a capacitor C59 (1nF 2kV 1206) and a resistor R57 (1M 1% 1206). The GND line is connected to the ground. The circuit also includes a diode D13 (Red 0805) and a resistor R20 (1K 1% 0603) for output regulation. The output voltage is labeled VBUS, and the output current is 3A. The circuit also includes a diode D12 (CMSH3-40MA or B340A-13-F) and a resistor R20 (1K 1% 0603) for output regulation.

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INPUT SUPPLY WITH REVERSE POLARITY PROTECTION

10V-36V to 5V 3A BUCK

This schematic diagram illustrates the input supply and buck converter circuit. The input is connected through a connector J9 (PJ-102AH) to a reverse polarity protection circuit consisting of a diode D11 (SMAZ15-13-F) and a MOSFET T1 (SQJ459EP-T1_GE3). The input voltage is labeled VIN, and the output of the protection circuit is labeled VIN_14V_PROT. The circuit includes several test points (TP8, TP9, TP7, TP1, TP2, TP3, TP4) and various passive components: resistors R13 (10K 1% 0805), R14 (100K 1% 0603), R15 (24.9K 1% 0603), R16 (100K 1% 0603), R17 (100K 1% 0603), R18 (40.2K 1% 0603), R19 (210K 1% 0603), R20 (1K 1% 0603), and R57 (1M 1% 1206). Capacitors include C19 (330uF 50V), C20 (10uF 50V 1206), C21 (100nF 50V 0603), C22 (150pF 50V 0603), C23 (100nF 50V 0603), C24 (10uF 50V 1206), C25 (10uF 50V 1206), C26 (10uF 50V 1206), C27 (10uF 50V 1206), C28 (10uF 50V 1206), and C59 (1nF 2kV 1206). The buck converter is implemented using the IC12 (MP4462DN-LF-Z) with pins VIN, EN, FREQ, COMP, GND, SW, and FB. The output of the buck converter is +5V, which is filtered by a capacitor C28 and a diode D12 (CMSH3-40MA or B340A-13-F). The output voltage is labeled +5V, and the output current is 3A. The circuit also includes a 15uH 4.2A inductor L1 and a 100K resistor R16 for the switching frequency.

USB CONNECTOR

This schematic diagram shows the USB connector circuit. The USB connector J8 (USB-B-S-F-B-TH) is connected to the VBUS, D+, D-, and SHIELD pins. The VBUS pin is connected to the VBUS line, which is filtered by a capacitor C17 (100uF 50V) and a diode D1 (Blue 0805). The D+ and D- pins are connected to the USB_D_P and USB_D_N lines, which are filtered by capacitors C26 (10uF 50V 1206) and C27 (10uF 50V 1206). The SHIELD pin is connected to the SHIELD line, which is filtered by a capacitor C59 (1nF 2kV 1206) and a resistor R57 (1M 1% 1206). The circuit also includes a diode D7 (SP0503BAHTG) and a resistor R41 (1K 1% 0402). The output of the USB connector is labeled USB_IND, and the output current is 1A.

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C							
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	Sheet Title: Revision History	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3
				Sheet: 6 of 6